

On the Independent Spanning Trees of Recursive Circulant Graphs $G(cd^m, d)$ with $d \geq 3$ *

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Abstract

Two spanning trees of a graph G are said to be independent if they are rooted at the same vertex r , and for each vertex $v \neq r$ in G , the two different paths from v to r , one path in each tree, are internally disjoint. A set of spanning trees of G is independent if they are pairwise independent. A recursive circulant graph $G(N, d)$ has $N = cd^m$ vertices labeled from 0 to $N - 1$, where $d \geq 2$, $m \geq 1$, and $1 \leq c < d$, and two vertices $x, y \in G(N, d)$ are adjacent if and only if there is an integer k with $0 \leq k \leq \lceil \log_d N \rceil - 1$ such that $x \pm d^k \equiv y \pmod{N}$. In this paper, we propose an algorithm to construct multiple independent spanning trees on recursive circulant graphs $G(cd^m, d)$ under the condition $d \geq 3$, where the number of independent spanning trees matches the connectivity of $G(cd^m, d)$.

Keywords: Recursive circulant graphs; Independent spanning trees; Internally disjoint paths; Fault-tolerant broadcasting;

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1 Introduction

Broadcasting is an important issue for numerous applications in parallel and distributed networks. In this paper, we deal with a fault-tolerant broadcasting problem, called *independent spanning trees problem* (ISTs problem for short), on a particular subclass of recursive circulant graphs.

For an undirected graph G , we denote the vertex set and the edge set of G by $V(G)$ and $E(G)$, respectively. For $x, y \in V(G)$, two paths P and Q joining x and y are said to be *internally disjoint*, denoted by $P \parallel Q$, if $E(P) \cap E(Q) = \emptyset$ and $V(P) \cap V(Q) = \{x, y\}$. A *rooted spanning tree* of a graph G is a subgraph of G which connects all vertices without forming cycles and provides a specified vertex as its root. If T is a tree and $x, y \in V(T)$, we denote $T[x, y]$ as the unique path from x to y in T . Two spanning trees T and T' of a graph G are said to be *independent* (refer as ISTs for short) if they are rooted at the same vertex, say r , and such that $T[r, x] \parallel T'[r, x]$ for every vertex $x \neq r$. We also refer a set of spanning trees of G to be independent if they

are pairwise independent.

The design of multiple ISTs has applications to the reliable communication protocols [3, 10, 16]. For example, a rooted spanning tree in the underlying graph of a network can be viewed as a broadcasting scheme for data communication. Thus, the fault tolerance can be achieved by sending k copies of the message along the k ISTs rooted at the source node. In recent years the problem of constructing multiple ISTs of a given graph has received much attention. However, this is a very hard problem for arbitrary graphs. In fact, Zehavi and Itai [28] conjectured that for any k -connected graph G and each vertex r of G , there exist k ISTs of G rooted at r . The conjecture has been confirmed only for k -connected graphs with $k \leq 4$ (see [10] for $k = 2$, [5, 28] for $k = 3$, and [7] for $k = 4$), and it is still open for arbitrary k -connected graphs when $k \geq 5$. For this reason, finding a set $\mathcal{T}$ of ISTs rooted at an arbitrary vertex of a graph G so that the cardinality of $\mathcal{T}$ matches the connectivity of G is therefore called the independent spanning trees problem of G . Till now, the ISTs problem has been solved for several restricted classes of interconnection networks, such as product graphs [16], chordal rings [11, 25], de-Bruijn and Kautz graphs [8, 9], and hypercubes [22, 26]. By providing the construction scheme, the ISTs problem on recursive circulant graphs is first considered in this paper.

A *circulant graph* $G(N, d)$, $d \geq 2$, is a simple graph (i.e., a graph without multiple edges and self-loop) with N vertices labeled from 0 to $N - 1$ and two vertices $x, y \in G(N, d)$ are adjacent if and only if there is an integer k with $0 \leq k \leq \lceil \log_d N \rceil - 1$ such that $x \pm d^k \equiv y \pmod{N}$. In particular, $G(N, d)$ is called a *recursive circulant graph* (or RC-graph for short) and possesses a recursive structure when $N = cd^m$, $1 \leq c < d$ and $m \geq 1$ [19, 20]. The existence of the parameter c enriches the class of RC-graphs to be a wider class of graphs such that the number of vertices of an RC-graph need not be the

power of d . Note that $G(cd^m, d)$ can be partitioned into d disjoint subgraphs such that each subgraph is isomorphic to $G(cd^{m-1}, d)$ and contains exactly those vertices with the same remainder of division by d . For example, Figure 1 shows the graph $G(16, 4)$. We can easily check that the set of vertices with the same remainder of division by 4 induces a subgraph isomorphic to $G(4, 4)$.

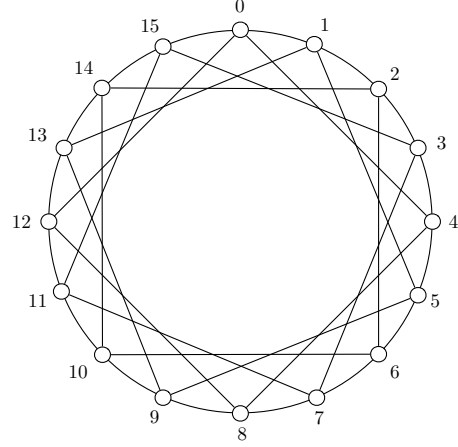


Figure 1: Example of recursive circulant graph $G(16, 4)$.

Due to the flexibility and extensibility of RC-graphs, this family of graphs was first introduced by Park and Chwa in [19] to serve as a topology of interconnection networks. Based on their structure properties, RC-graphs are suitable for developing algorithms, such as routing algorithms [6, 13, 20, 21] and embedding schemes [12, 14, 20, 21]. Besides, RC-graphs are also easy for analyzing network metrics, such as diameter [20], bisection width [27], connectivity [20, 24] and hamiltonian-like properties [1, 2, 4, 15, 17, 18, 20, 23].

As it is, $G(cd^m, d)$ is defined to be a Cayley graph over the cyclic group Z_N with generating set $\{d^0, d^1, \dots, d^{\lceil \log_d N \rceil - 1}\}$, where d^k is called the k th *jump*. It is well-known that every Cayley graph over a general group is vertex-symmetric, and thus is regular. Let δ_m denote the degree of vertices in $G(cd^m, d)$.

Then, $\delta_m = \delta_{m-1} + 2$ for $m \geq 1$. In [20], the closed form of δ_m is given as follows:

$$\delta_m = \begin{cases} 2m-1, & \text{if } c=1 \text{ and } d=2; \\ 2m, & \text{if } c=1 \text{ and } d>2; \\ 2m+1, & \text{if } c=2; \\ 2m+2, & \text{if } c>2. \end{cases}$$

Furthermore, Park and Chwa [20] have pointed out that $\kappa_m = \lambda_m = \delta_m$, where κ_m and λ_m denote the connectivity and the edge-connectivity of $G(cd^m, d)$, respectively. In addition, they also pointed out that RC-graphs are closely related to some interesting classes of interconnection networks. For instance, $G(2^m, 2)$ is a supergraph of an m -dimensional hypercube Q_m , $G(2^m, 4)$ has the same number of nodes and edges as Q_m , and every $G(2^m, 2^k)$ with $2 \leq k < m$ is a tripartite graph. In particular, many researches have been focused on the subclass $G(2^m, 4)$ (e.g., see [13–15, 18, 24, 27]). In this paper, we deal with ISTs problem on $G(cd^m, d)$ with $d \geq 3$, which forms a superclass of $G(2^m, 4)$.

We organize the remaining part of this paper as follows. Section 2 introduces a useful concept called path-decomposition Latin square (PDLS) and gives some fundamental properties which will facilitate our discussion. Section 3 presents our algorithm that can construct δ_m ISTs on $G(cd^m, d)$ with $d \geq 3$. In Section 4, we show the correctness of our algorithm. The last section contains our concluding remarks.

2 Preliminaries

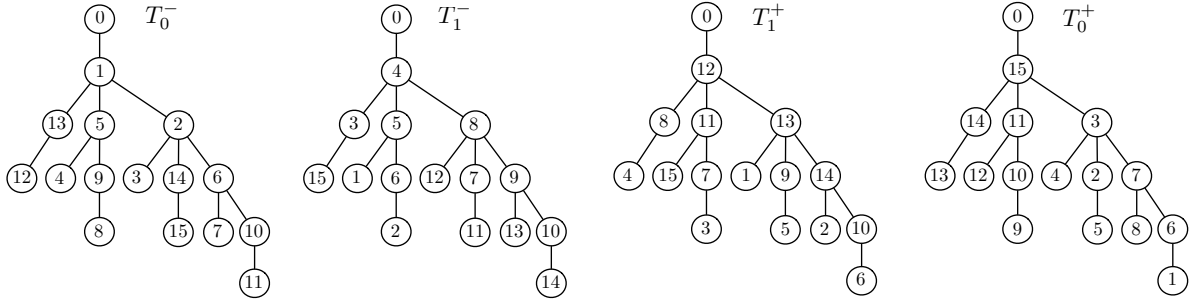
From now on, unless otherwise mentioned, all arithmetics are taken modulo $N(= cd^m)$ using the appropriate residues. We first give some definitions and notations which are necessary for illustrating our algorithm.

The *neighborhood* of a vertex x , denoted by $N(x)$, is the set of vertices adjacent to x in a graph. Since RC-graphs are vertex-symmetric, without loss of generality, we sim-

ply consider vertex 0 as the root of ISTs for $G(cd^m, d)$. Obviously, $N(0) = \{\pm d^k : 0 \leq k \leq \lceil \log_d N \rceil - 1\}$. Suppose that $\mathcal{T}$ is a set consisting of δ_m ISTs rooted at 0 for $G(cd^m, d)$. For each $T \in \mathcal{T}$, the *parent* of a vertex $x (\neq 0)$, denoted by $\text{parent}(T, x)$, is a vertex adjacent to x in the path $T[x, 0]$. From the connectivity of $G(cd^m, d)$ and the fact $|\mathcal{T}| = \delta_m$, there is only one child of 0 in every spanning tree $T \in \mathcal{T}$. Thus, if $x = +d^k$ (respectively, $x = -d^k$), then it uses the jump $j = -d^k$ (respectively, $j = +d^k$) to connect the root. Moreover, $j = -d^k$ (respectively, $j = +d^k$) is the last jump in a path connecting every vertex and the root. Therefore, we denote by T_k^- (respectively, T_k^+) for specifying the IST with the root taking $+d^k$ (respectively, $-d^k$) as its child. For example, four ISTs rooted at vertex 0 for $G(16, 4)$ are shown in Figure 2.

To explicitly represent the adjacency, we use the notation $x \xrightarrow{+d^k} y$ if $x + d^k = y$ (respectively, $x \xrightarrow{-d^k} y$ if $x - d^k = y$) to mean that both vertices $x, y \in G(cd^m, d)$ are connected by the k th jump. Hence, we can represent a path by using this notation repeatedly. Let $J = \{\pm d^k : 0 \leq k \leq \lceil \log_d N \rceil - 1\}$. If a jump $j \in J$ occurs successively to connect x and y , then we use the notation $x \xrightarrow{j/\alpha} y$ to mean $y = x + \alpha \cdot j$, where α is the number of repetitions of j between x and y . Let $x (\neq 0) \in G(cd^m, d)$ be any vertex and suppose that $x_0 \xrightarrow{j_1} x_1 \xrightarrow{j_2} x_2 \xrightarrow{j_3} \dots \xrightarrow{j_{\ell-1}} x_{\ell-1} \xrightarrow{j_\ell} x_\ell$ is a shortest path joining $x (= x_0)$ and $0 (= x_\ell)$, where each $j_i \in J$ is a jump. As a matter of convenience, such a path is called a shortest path of x and is denoted by P_x . Since x can be decomposed into the sum $\sum_{i=1}^{\ell} (-j_i)$, we say a *decomposition* of x to mean the terms for those jumps. For $j \in J$, we denote by $n(P_x, j)$ the number of occurrences of j appeared in P_x . Since P_x is a shortest path, the following conditions hold:

- (i) $x_i \neq x_k$ for $0 \leq i, k \leq \ell$ and $i \neq k$;
- (ii) $j_i \neq -j_k$ for $1 \leq i, k \leq \ell$ and $i \neq k$; and
- (iii) $n(P_x, j) \leq \lfloor \frac{d}{2} \rfloor$ for every $j \in J$.


 Figure 2: A set of independent spanning trees on $G(16, 4)$.

A *Latin square* is a square matrix with n^2 entries chosen from a set H of n distinct elements such that none of the elements occurs twice within any row or column of the matrix. A *path-decomposition Latin square* (PDLS for short) with respect to a shortest path P_x is a Latin square whose entries are chosen from the set $H_{P_x} = \{j \in J : n(P_x, j) \neq 0\}$. In particular, we define the following matrix.

Definition 1. Suppose that $H_{P_x} = \{j_0, j_1, \dots, j_{t-1}\}$ with $|j_0| < |j_1| < \dots < |j_{t-1}|$. Then, the matrix defined below is called the *increasingly rotational PDLS* of P_x :

$$I(P_x) = \begin{bmatrix} j_0 & j_1 & \dots & j_{t-2} & j_{t-1} \\ j_1 & j_2 & \dots & j_{t-1} & j_0 \\ \vdots & & \dots & & \\ j_{t-2} & j_{t-1} & \dots & j_{t-4} & j_{t-3} \\ j_{t-1} & j_0 & \dots & j_{t-3} & j_{t-2} \end{bmatrix}.$$

In addition, the entry j_{i+1} in $I(P_x)$ is called the *successor* of j_i and is denoted by $\text{succ}(I(P_x), j_i)$, where the indices i and $i+1$ are taken modulo t . (Hereafter, all arithmetics applied to the indices of the entries in PDLS are taken modulo $|H_{P_x}|$.)

Definition 2. For a shortest path P_x , we let $r = \sum_{j \in H_{P_x}} n(P_x, j)$. A *PDLS of P_x with repetitions*, denoted by $I^*(P_x)$, is a matrix of size $t \times r$ obtained from $I(P_x)$ such that every entry j in a row of $I(P_x)$ is substituted by $n(P_x, j)$ repetitions of j in the same row of $I^*(P_x)$.

Because each row of $I^*(P_x)$ can be viewed as an integer sequence, we also define the following sequences:

Definition 3. For a shortest path P_x , a *regular sequence* R of P_x , denoted by $R = [P_x]$, is an integer sequence chosen from a row of $I^*(P_x)$. In particular, we denote by $R = [P_x]_j$ to mean a regular sequence R that contains a specific jump $j \in H_{P_x}$ as its last element. Moreover, an integer sequence S is called an *extended sequence* of P_x , denoted by $S = (\alpha \cdot j') || [P_x]$, if S is obtained from $R = [P_x]$ by adding α repetitions of a jump $j' \in J$ in the front of R .

Definition 4. Let $A = a_1, a_2, \dots, a_t$ be a sequence of integers. A *prefix-sum* of A is defined to be a sum $\sum_{i=1}^k a_i$ for some $k = 1, 2, \dots, t$. In particular, a *strict prefix-sum* is a prefix-sum satisfying $k \neq t$. Also, define $\mathcal{S}_A = \{\sum_{i=1}^k a_i : k = 1, 2, \dots, t-1\}$ as the set containing all strict prefix-sums of A .

Example 1. We consider the graph $G(4 \times 5^3, 5)$ and a shortest path $P = (327 \xrightarrow{-5^0} 326 \xrightarrow{-5^0} 325 \xrightarrow{+5^2} 350 \xrightarrow{+5^2} 375 \xrightarrow{+5^3} 0)$. The two particular matrices defined above are given by

$$I(P) = \begin{bmatrix} -5^0 & +5^2 & +5^3 \\ +5^2 & +5^3 & -5^0 \\ +5^3 & -5^0 & +5^2 \end{bmatrix}$$

and

$$I^*(P) = \begin{bmatrix} -5^0 & -5^0 & +5^2 & +5^2 & +5^3 \\ +5^2 & +5^2 & +5^3 & -5^0 & -5^0 \\ +5^3 & -5^0 & -5^0 & +5^2 & +5^2 \end{bmatrix}.$$

The strict prefix-sums for each regular sequence of P are the following:

$$\begin{aligned}\mathcal{S}_{[P]_{+5^3}} &= \{-1(\equiv 499), -2(\equiv 498), 23, 48\}, \\ \mathcal{S}_{[P]_{-5^0}} &= \{25, 50, 175, 174\}, \\ \mathcal{S}_{[P]_{+5^2}} &= \{125, 124, 123, 148\}.\end{aligned}$$

Besides, the strict prefix-sums of two extended sequences $S_1 = (2 \cdot -5^0)[P]_{-5^0}$ and $S_2 = (-5^3)[P]_{+5^3}$ are the following:

$$\begin{aligned}\mathcal{S}_{S_1} &= \{-1(\equiv 499), -2(\equiv 498), 23, 48, 173, 172\}, \\ \mathcal{S}_{S_2} &= \{-125(\equiv 375), 374, 373, 398, 423\}.\end{aligned}\quad \square$$

Due to space limitations, we omit the proofs in following propositions.

Proposition 1. *For a shortest path P_x , if R and R' are two distinct regular sequences of P_x , then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

To demonstrate more properties related to PDLS with respect to a shortest path P_x , we also define $H_{\overline{P_x}} = \{j \in J : -j \in H_{P_x}\}$.

Proposition 2. *Let $x, y (\neq 0) \in G(cd^m, d)$ be two vertices such that $y = x + \alpha \cdot j$ for some jump $j \in H_{\overline{P_x}} \cap H_{P_y}$, where $1 \leq \alpha \leq d - n(P_x, -j) - n(P_y, j)$. If $R = [P_x]$ and $R' = (\alpha \cdot j)[P_y]_j$, then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

Proposition 3. *Let $x, y (\neq 0) \in G(cd^m, d)$ be two vertices such that $y = x - j$ for some jump $j \in H_{P_y} \setminus (H_{P_x} \cup H_{\overline{P_x}})$. If $R = [P_x]$ and $R' = (-j)[P_y]_j$, then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

Proposition 4. *Let $x, y, z (\neq 0) \in G(cd^m, d)$ such that $y = x + \alpha \cdot j$ and $z = x + \beta \cdot j'$ are two distinct vertices for some jumps $j \in H_{P_y} \cap H_{\overline{P_z}}$ and $j' \in H_{P_z}$, where $1 \leq \alpha \leq d - n(P_z, -j) - n(P_y, j)$ and $\beta \geq 1$. If $R = (\alpha \cdot j)[P_y]_j$ and $R' = (\beta \cdot j')[P_z]_{j'}$, then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

Proposition 5. *Let $x, y, z (\neq 0) \in G(cd^m, d)$ such that $y = x - j$ and $z = x - j'$ are two distinct vertices for some jumps $j \in H_{P_y} \setminus (H_{P_z} \cup H_{\overline{P_z}})$ and $j' \in H_{P_z} \setminus (H_{P_y} \cup H_{\overline{P_y}})$. If $R = (-j)[P_y]_j$ and $R' = (-j')[P_z]_{j'}$, then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

Proposition 6. *Let $x, y, z (\neq 0) \in G(cd^m, d)$ such that $y = x + \alpha \cdot j$ and $z = x - j'$ are two distinct vertices for some jumps $j \in H_{P_y} \cap H_{\overline{P_z}}$ and $j' \in H_{P_z}$, where $1 \leq \alpha \leq d - n(P_z, -j) - n(P_y, j)$. If $R = (\alpha \cdot j)[P_y]_j$ and $R' = (-j')[P_z]_{j'}$, then $\mathcal{S}_R \cap \mathcal{S}_{R'} = \emptyset$.*

3 Constructing ISTs on $G(cd^m, d)$ with $d \geq 3$

In this section, we present our algorithm for solving ISTs problem on RC-graphs with $d \geq 3$. Let x be any vertex in $G(cd^m, d)$ and suppose that $x = (x_m x_{m-1} \cdots x_1 x_0)_d$ is represented as an integer with base d . Then,

$$x = x_m d^m + x_{m-1} d^{m-1} + \cdots + x_1 d + x_0.$$

The following procedure called SHORTEST-PATH will be used to find a shortest path P_x (i.e., a shortest path between x and 0). As a replacement for listing the vertices of P_x , we establish the set H_{P_x} . To simplify the notations in the algorithm, we write H instead of H_{P_x} , and $n(j)$ instead of $n(P_x, j)$ for each jump $j \in J$.

Procedure SHORTEST-PATH(x)

begin

$H = \emptyset$; Carry = 0;

For $i = 0$ to $m - 1$ do

if $0 < x_i \leq \lfloor d/2 \rfloor$

$H = H \cup \{-d^i\}$; $n(-d^i) = x_i$;

$n(+d^i) = 0$; Carry = 0;

else if $\lfloor d/2 \rfloor < x_i \leq d - 1$

$H = H \cup \{+d^i\}$;

$n(+d^i) = d - x_i$; $n(-d^i) = 0$;

$x_{i+1} = x_{i+1} + 1$; Carry = 1;

else

$n(-d^i) = n(+d^i) = 0$;

if Carry = 1

$x_{i+1} = x_{i+1} + 1$;

endif

enddo

if $c \geq 2$ and $0 < x_m \leq \lfloor c/2 \rfloor$

$H = H \cup \{-d^m\}$; $n(-d^m) = x_m$;

```

     $n(+d^m) = 0;$ 
  else if  $c > 2$  and  $\lfloor c/2 \rfloor < x_m \leq c - 1$ 
     $H = H \cup \{+d^m\};$ 
     $n(+d^m) = c - x_m; \quad n(-d^m) = 0;$ 
  else
     $n(-d^m) = n(+d^m) = 0;$ 
  endif
end SHORTEST-PATH

```

As a result, a shortest path P_x can be described by using the jumps on each row in the matrix $I^*(P_x)$. For instance, Example 1 shows that $P = (327 \xrightarrow{-5^0} 326 \xrightarrow{-5^0} 325 \xrightarrow{+5^2} 350 \xrightarrow{+5^2} 375 \xrightarrow{+5^3} 0)$ is a shortest path of $327 = (2302)_5$ in $G(4 \times 5^3, 5)$.

Let $\text{sign}(i)$ be defined as follows:

$$\text{sign}(i) = \begin{cases} - & \text{if } 2i \leq \delta_m - 1 \\ + & \text{if } 2i > \delta_m - 1. \end{cases} \quad (1)$$

Using a set of simple rules referred to H_{P_x} and $H_{\overline{P_x}}$, we can determine the parent of x in each spanning tree $T \in \mathcal{T}$, where $\mathcal{T}$ is the set of all ISTs constructed by the algorithm. For each vertex $x(\neq 0) \in G(cd^m, d)$, we perform the following procedure:

Procedure GEN-PARENTS(x)

begin

Determine H_{P_x} , $H_{\overline{P_x}}$, and $J \setminus (H_{P_x} \cup H_{\overline{P_x}})$ for x ;

for $i = 0$ to $\delta_m - 1$ **do**

$j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}};$

$\text{parent}(T_{\min\{i, \delta_m - 1 - i\}}^{\text{sign}(i)}, x) =$

$$\begin{cases} x + \text{succ}(I(P_x), j) & \text{if } j \in H_{P_x}; \\ x + j & \text{if } j \in H_{\overline{P_x}}; \\ x - j & \text{otherwise;} \end{cases} \quad (2)$$

enddo

end GEN-PARENTS

Example 2. We again consider $G(4 \times 5^3, 5)$ as an example. Table 1 shows three types of jumps for every vertex $x \in N(327) \cup \{327\}$

in $G(4 \times 5^3, 5)$. According to (2), all informations for calculating the parents of vertices 327, 302, and 332 in every spanning tree are summarized in Table 2. From this table, we can see that 332 (respectively, 302) is the parent of 327 in T_1^- (respectively, in T_2^-).

4 Correctness

In what follows, we will prove the correctness of our algorithm. For convenience, we simply write T_i , $0 \leq i \leq \delta_m - 1$, to denote the graph $T_{\min\{i, \delta_m - 1 - i\}}^{\text{sign}(i)}$ and let $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}}$. Again, proofs of the following lemmas are omitted due to space constraints. The first two lemmas show that if $j \notin H_{P_x}$, then there is a vertex $y \in T_i[x, 0]$ such that $j \in H_{P_y}$.

Lemma 7. *Let $x(\neq 0)$ be any vertex in $G(cd^m, d)$. If $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}} \in H_{\overline{P_x}}$ for some $0 \leq i \leq \delta_m - 1$, then there exists a vertex $y = x + \alpha \cdot j$ such that $\alpha = d - n(P_x, -j) - n(P_y, j)$ and $j \in H_{P_y}$.*

Lemma 8. *Let $x(\neq 0)$ be any vertex in $G(cd^m, d)$. If $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}} \notin (H_{P_x} \cup H_{\overline{P_x}})$ for some $0 \leq i \leq \delta_m - 1$, then there exists a vertex $y = x - j$ (i.e., $y = \text{parent}(T_i, x)$) such that $j \in H_{P_y}$. In particular, $H_{P_y} = H_{P_x} \cup \{j\}$.*

Lemma 9. *Let $x(\neq 0)$ be any vertex in $G(cd^m, d)$ and suppose that $j, j' \in H_{\overline{P_x}}$, where $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}}$ and $j' = \text{sign}(i') \times d^{\min\{i', \delta_m - 1 - i'\}}$ for some $0 \leq i, i' \leq \delta_m - 1$ and $i \neq i'$. If $y = x + \alpha \cdot j$ and $z = x + \beta \cdot j'$ are two vertices in $G(cd^m, d)$ and $|j| < |j'|$, then $j \in H_{\overline{P_z}}$ and $\alpha + n(P_z, -j) + n(P_y, j) = d$.*

Lemma 10. *Let $x(\neq 0)$ be any vertex in $G(cd^m, d)$ and suppose that $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}} \in H_{\overline{P_x}}$ and $j' = \text{sign}(i') \times d^{\min\{i', \delta_m - 1 - i'\}} \notin (H_{P_x} \cup H_{\overline{P_x}})$ for some $0 \leq i, i' \leq \delta_m - 1$. If $y = x + \alpha \cdot j$ and $z = x - j'$ are two vertices in $G(cd^m, d)$, then $j \in H_{\overline{P_z}}$ and $\alpha + n(P_z, -j) + n(P_y, j) = d$.*

Table 1: Three types of jumps for each vertex $x \in N(327) \cup \{327\}$ in $G(4 \times 5^3, 5)$

x	H_{P_x}	$H_{\overline{P_x}}$	$J \setminus (H_{P_x} \cup H_{\overline{P_x}})$
$327 = (2302)_5$	$\{-5^0, +5^2, +5^3\}$	$\{+5^0, -5^2, -5^3\}$	$\{-5^1, +5^1\}$
$202 = 327 - 5^3 = (1302)_5$	$\{-5^0, +5^2, -5^3\}$	$\{+5^0, -5^2, +5^3\}$	$\{-5^1, +5^1\}$
$302 = 327 - 5^2 = (2202)_5$	$\{-5^0, -5^2, -5^3\}$	$\{+5^0, +5^2, +5^3\}$	$\{-5^1, +5^1\}$
$322 = 327 - 5^1 = (2242)_5$	$\{-5^0, +5^1, +5^2, +5^3\}$	$\{+5^0, -5^1, -5^2, -5^3\}$	$\emptyset$
$326 = 327 - 5^0 = (2301)_5$	$\{-5^0, +5^2, +5^3\}$	$\{+5^0, -5^2, -5^3\}$	$\{-5^1, +5^1\}$
$328 = 327 + 5^0 = (2303)_5$	$\{+5^0, -5^1, +5^2, +5^3\}$	$\{-5^0, +5^1, -5^2, -5^3\}$	$\emptyset$
$332 = 327 + 5^1 = (2312)_5$	$\{-5^0, -5^1, +5^2, +5^3\}$	$\{+5^0, +5^1, -5^2, -5^3\}$	$\emptyset$
$352 = 327 + 5^2 = (2402)_5$	$\{-5^0, +5^2, +5^3\}$	$\{+5^0, -5^2, -5^3\}$	$\{-5^1, +5^1\}$
$452 = 327 + 5^3 = (3302)_5$	$\{-5^0, +5^2\}$	$\{+5^0, -5^2\}$	$\{-5^1, +5^1, -5^3, +5^3\}$

Table 2: The parents of vertices 327, 302, and 332 in each IST of $G(4 \times 5^3, 5)$

x	i	$\min\{i, \delta_m - 1 - i\}$	the condition for the jump j	IST	parent of x
327	0	0	$j = -5^0 \in H_{P_x}$	T_0^-	$327 + 5^2 = 352$
	1	1	$j = -5^1 \in J \setminus (H_{P_x} \cup H_{\overline{P_x}})$	T_1^-	$327 + 5^1 = 332$
	2	2	$j = -5^2 \in H_{\overline{P_x}}$	T_2^-	$327 - 5^2 = 302$
	3	3	$j = -5^3 \in H_{\overline{P_x}}$	T_3^-	$327 - 5^3 = 202$
	4	3	$j = +5^3 \in H_{P_x}$	T_3^+	$327 - 5^0 = 326$
	5	2	$j = +5^2 \in H_{P_x}$	T_2^+	$327 + 5^3 = 452$
	6	1	$j = +5^1 \in J \setminus (H_{P_x} \cup H_{\overline{P_x}})$	T_1^+	$327 - 5^1 = 322$
	7	0	$j = +5^0 \in H_{\overline{P_x}}$	T_0^+	$327 + 5^0 = 328$
332	0	0	$j = -5^0 \in H_{P_x}$	T_0^-	$332 - 5^1 = 327$
	1	1	$j = -5^1 \in H_{P_x}$	T_1^-	$332 + 5^2 = 357$
	2	2	$j = -5^2 \in H_{\overline{P_x}}$	T_2^-	$332 - 5^2 = 307$
	3	3	$j = -5^3 \in H_{\overline{P_x}}$	T_3^-	$332 - 5^3 = 207$
	4	3	$j = +5^3 \in H_{P_x}$	T_3^+	$332 - 5^0 = 331$
	5	2	$j = +5^2 \in H_{P_x}$	T_2^+	$332 + 5^3 = 457$
	6	1	$j = +5^1 \in H_{\overline{P_x}}$	T_1^+	$332 + 5^1 = 337$
	7	0	$j = +5^0 \in H_{\overline{P_x}}$	T_0^+	$332 + 5^0 = 333$
302	0	0	$j = -5^0 \in H_{P_x}$	T_0^-	$302 - 5^2 = 277$
	1	1	$j = -5^1 \in J \setminus (H_{P_x} \cup H_{\overline{P_x}})$	T_1^-	$302 + 5^1 = 307$
	2	2	$j = -5^2 \in H_{P_x}$	T_2^-	$302 - 5^3 = 177$
	3	3	$j = -5^3 \in H_{P_x}$	T_3^-	$302 - 5^0 = 301$
	4	3	$j = +5^3 \in H_{\overline{P_x}}$	T_3^+	$302 + 5^3 = 427$
	5	2	$j = +5^2 \in H_{\overline{P_x}}$	T_2^+	$302 + 5^2 = 327$
	6	1	$j = +5^1 \in J \setminus (H_{P_x} \cup H_{\overline{P_x}})$	T_1^+	$302 - 5^1 = 297$
	7	0	$j = +5^0 \in H_{\overline{P_x}}$	T_0^+	$302 + 5^0 = 303$

The following lemma shows that every subgraph constructed in $\mathcal{T}$ is a spanning tree.

Lemma 11. *For every integer i with $0 \leq i \leq \delta_m - 1$, T_i is a spanning tree rooted at 0 in $G(cd^m, d)$.*

Proof. Let $i' = \delta_m - 1 - i$ and $j = \text{sign}(i) \times d^{\min\{i, i'\}}$. It is obvious from Procedure GEN-PARENTS that every vertex $x (\neq 0) \in G(cd^m, d)$

implies $x \in T_i$. Thus the resulting graph is a spanning subgraph of $G(cd^m, d)$. To complete the proof, we need to show that there exists a unique path from every vertex $x (\neq 0)$ to 0 in T_i . In particular, the vertex $-j$ is adjacent to 0 in T_i . Suppose that $H_{P_x} = \{j_0, j_1, \dots, j_{s-1}\}$ with $|j_0| < |j_1| < \dots < |j_{s-1}|$. Then, each $n(P_x, j_k)$, $k = 0, 1, \dots, s-1$, in the shortest path P_x can be calculated by Procedure

SHORTEST-PATH. According to j , we consider the following cases:

Case 1: $j \in H_{P_x}$. Suppose $j = j_p$ for some $p = 0, 1, \dots, s-1$. Since $j_{p+1} = \text{succ}(I(P_x), j_p)$, x is adjacent to $x + j_{p+1}$ in T_i by (2). Let $y = x + n(P_x, j_{p+1}) \cdot j_{p+1}$. Clearly, P_y is a subpath of P_x and $H_{P_y} = \{j_0, j_1, \dots, j_p, j_{p+2}, \dots, j_{s-1}\}$. Thus, $n(P_y, j_k) = n(P_x, j_k)$ for every $j_k \in H_{P_y}$. Since $j_p \in H_{P_y}$ and $j_{p+2} = \text{succ}(I(P_y), j_p)$, y is adjacent to $y + j_{p+2}$ in T_i by (2). Let $z = y + n(P_y, j_{p+2}) \cdot j_{p+2}$. Clearly, P_z is a subpath of P_y and $H_{P_z} = \{j_0, j_1, \dots, j_p, j_{p+3}, \dots, j_{s-1}\}$. Thus, $n(P_z, j_k) = n(P_y, j_k)$ for every $j_k \in H_{P_z}$. Again by (2), z is adjacent to $z + j_{p+3}$ in T_i . By this way, we can find the following unique path connecting x and 0 in T_i :

$$\begin{aligned} T_i[x, 0] : & x \xrightarrow[n(P_x, j_{p+1})]{j_{p+1}} \left(x + n(P_x, j_{p+1}) \cdot j_{p+1} \right) \\ & \xrightarrow[n(P_x, j_{p+2})]{j_{p+2}} \left(x + n(P_x, j_{p+1}) \cdot j_{p+1} + n(P_x, j_{p+2}) \cdot j_{p+2} \right) \\ & \xrightarrow[n(P_x, j_{p+3})]{j_{p+3}} \dots \dots \dots \xrightarrow[n(P_x, j_{p-1})]{j_{p-1}} \left(-n(P_x, j_p) \cdot j_p \right) \\ & \xrightarrow[n(P_x, j_p)]{j_p} 0. \end{aligned}$$

Case 2: $j \in H_{\bar{P}_x}$. Suppose x has the representation $(x_m x_{m-1} \dots x_1 x_0)_d$. By (2), x is adjacent to $x + j$ in T_i . Let

$$\alpha = \begin{cases} x_i - \lfloor \frac{d}{2} \rfloor & \text{if } j < 0 \\ \lfloor \frac{d}{2} \rfloor + 1 - x_{i'} & \text{if } j > 0 \end{cases}$$

and let $y = x + \alpha \cdot j$. By Lemma 7, $j \in H_{P_y}$. Also, we have shown in Case 1 that there exists a unique path $T_i[y, 0]$ that connects y and 0 in T_i . Therefore, $x \xrightarrow[\alpha]{j} y$ along with the path $T_i[y, 0]$ forms the unique path connecting x and 0 in T_i .

Case 3: $j \in J \setminus (H_{P_x} \cup H_{\bar{P}_x})$. By (2), x is adjacent to $x - j$ in T_i . Let $y = x - j$. By Lemma 8, $j \in H_{P_y}$. Also, we have shown in Case 1 that there exists a unique path $T_i[y, 0]$ that connects y and 0 in T_i . Therefore, $x \xrightarrow{-j} y$ along with the path $T_i[y, 0]$ forms the unique path connecting x and 0 in T_i . $\square$

We now show the independence of ISTs in $\mathcal{T}$.

Lemma 12. For $0 \leq i, i' \leq \delta_m - 1$ and $i \neq i'$, T_i and $T_{i'}$ are ISTs rooted at 0 in $G(cd^m, d)$.

Proof. Let $x (\neq 0)$ be any vertex in $G(cd^m, d)$ and let $j = \text{sign}(i) \times d^{\min\{i, \delta_m - 1 - i\}}$ and $j' = \text{sign}(i') \times d^{\min\{i', \delta_m - 1 - i'\}}$. We consider the following cases:

Case 1: $j, j' \in H_{P_x}$. $T_i[x, 0] \parallel T_{i'}[x, 0]$ directly follows from Proposition 1.

Case 2: $j \in H_{P_x}$ and $j' \in H_{\bar{P}_x}$. By Lemma 11, $T_i[x, 0]$ takes j as its last jump and there is a vertex $y = x + \alpha \cdot j'$ such that $x \xrightarrow[\alpha]{j'} y$ along with the path $T_{i'}[y, 0]$ forms the unique path connecting x and 0 in $T_{i'}$. In particular, $T_{i'}[y, 0]$ takes j' as its last jump. By Lemma 7, we have $j' \in H_{P_y}$ and $\alpha = d - n(P_x, -j') - n(P_y, j')$. Let $R = [P_x]_j$ and $R' = (\alpha \cdot j')|[P_y]_{j'}$. Since $j' \in H_{\bar{P}_x} \cap H_{P_y}$, by Proposition 2, if $s_1 \in \mathcal{S}_R$ and $s_2 \in \mathcal{S}_{R'}$, then $s_1 \neq s_2$. Thus, $T_i[x, 0] \parallel T_{i'}[x, 0]$.

Case 3: $j \in H_{P_x}$ and $j' \in J \setminus (H_{P_x} \cup H_{\bar{P}_x})$. By Lemma 11, $T_i[x, 0]$ takes j as its last jump and there is a vertex $y = x - j'$ such that $x \xrightarrow{-j'} y$ along with the path $T_{i'}[y, 0]$ forms the unique path connecting x and 0 in $T_{i'}$. In particular, $T_{i'}[y, 0]$ takes j' as its last jump. Thus, $j' \in H_{P_y}$. Let $R = [P_x]_j$ and $R' = (-j')|[P_y]_{j'}$. Since $j' \notin H_{P_x} \cup H_{\bar{P}_x}$ and $j' \in H_{P_y}$, by Proposition 3, if $s_1 \in \mathcal{S}_R$ and $s_2 \in \mathcal{S}_{R'}$, then $s_1 \neq s_2$. Thus, $T_i[x, 0] \parallel T_{i'}[x, 0]$.

Case 4: $j, j' \in H_{\bar{P}_x}$. By Lemma 11, there is a vertex $y = x + \alpha \cdot j$ such that $x \xrightarrow[\alpha]{j} y$ along with the path $T_i[y, 0]$ forms the unique path connecting x and 0 in T_i . Similarly, there is a vertex $z = x + \beta \cdot j'$ such that $x \xrightarrow[\beta]{j'} z$ along with the path $T_{i'}[z, 0]$ forms the unique path connecting x and 0 in $T_{i'}$. In particular, $T_i[y, 0]$ takes j as its last jump and $T_{i'}[z, 0]$ takes j' as its last jump. Thus, $j \in H_{P_y}$ and $j' \in H_{P_z}$. Without loss of generality, we assume $|j| < |j'|$. By Lemma 9, $j \in H_{\bar{P}_z}$ and $\alpha + n(P_z, -j) + n(P_y, j) = d$. Let $R = (\alpha \cdot j)|[P_y]_j$ and $R' = (\beta \cdot j')|[P_z]_{j'}$. Since $j \in H_{P_y} \cap H_{\bar{P}_z}$ and $j' \in H_{P_z}$, by Proposi-

tion 4, if $s_1 \in \mathcal{S}_R$ and $s_2 \in \mathcal{S}_{R'}$, then $s_1 \neq s_2$. Thus, $T_i[x, 0] \parallel T_{i'}[x, 0]$.

Case 5: $j, j' \in J \setminus (H_{P_x} \cup H_{\overline{P}_x})$. By Lemma 11, there is a vertex $y = x - j$ such that $x \xrightarrow{-j} y$ along with the path $T_i[y, 0]$ forms the unique path connecting x and 0 in T_i . Similarly, there is a vertex $z = x - j'$ such that $x \xrightarrow{-j'} z$ along with the path $T_{i'}[z, 0]$ forms the unique path connecting x and 0 in $T_{i'}$. In particular, $T_i[y, 0]$ takes j as its last jump and $T_{i'}[z, 0]$ takes j' as its last jump. Thus, $j \in H_{P_y}$ and $j' \in H_{P_z}$. Since $j, j' \notin H_{P_x} \cup H_{\overline{P}_x}$, none of the terms $j, -j, j'$ and $-j'$ are contained in H_{P_x} or $H_{\overline{P}_x}$. By Lemma 8, $H_{P_y} = H_{P_x} \cup \{j\}$ and $H_{P_z} = H_{P_x} \cup \{j'\}$. This shows that $j \notin H_{P_z} \cup H_{\overline{P}_z}$ and $j' \notin H_{P_y} \cup H_{\overline{P}_y}$. Let $R = (-j)[P_y]_j$ and $R' = (-j')[P_z]_{j'}$. Since $j \in H_{P_y} \setminus (H_{P_z} \cup H_{\overline{P}_z})$ and $j' \in H_{P_z} \setminus (H_{P_y} \cup H_{\overline{P}_y})$, by Proposition 5, if $s_1 \in \mathcal{S}_R$ and $s_2 \in \mathcal{S}_{R'}$, then $s_1 \neq s_2$. Thus, $T_i[x, 0] \parallel T_{i'}[x, 0]$.

Case 6: $j \in H_{\overline{P}_x}$ and $j' \in J \setminus (H_{P_x} \cup H_{\overline{P}_x})$. By Lemma 11, there is a vertex $y = x + \alpha \cdot j$ such that $x \xrightarrow{\alpha \cdot j} y$ along with the path $T_i[y, 0]$ forms the unique path connecting x and 0 in T_i . Also, there is a vertex $z = x - j'$ such that $x \xrightarrow{-j'} z$ along with the path $T_{i'}[z, 0]$ forms the unique path connecting x and 0 in $T_{i'}$. In particular, $T_i[y, 0]$ takes j as its last jump and $T_{i'}[z, 0]$ takes j' as its last jump. Thus, $j \in H_{P_y}$ and $j' \in H_{P_z}$. By Lemma 10, $j \in H_{\overline{P}_z}$ and $\alpha + n(P_z, -j) + n(P_y, j) = d$. Let $R = (\alpha \cdot j)[P_y]_j$ and $R' = (-j')[P_z]_{j'}$. Since $j \in H_{P_y} \cap H_{\overline{P}_z}$ and $j' \in H_{P_z}$, by Proposition 6, if $s_1 \in \mathcal{S}_R$ and $s_2 \in \mathcal{S}_{R'}$, then $s_1 \neq s_2$. Thus, $T_i[x, 0] \parallel T_{i'}[x, 0]$.

Since we choose x arbitrarily from $G(cd^m, d)$, T_i and $T_{i'}$ are independent. $\square$

Combining Lemma 11 and Lemma 12, we have the following theorem.

Theorem 13. *The algorithm of constructing δ_m ISTs in $G(cd^m, d)$ with $d \geq 3$ can be executed in $O(mN)$ time, where $N = c \times d^m$. In particular, the algorithm can be parallelized on*

$G(cd^m, d)$ by using $O(N)$ processors to run in $O(m)$ time.

5 Concluding Remarks

In this paper, we present an algorithm for constructing δ_m ISTs on RC-graphs $G(cd^m, d)$ with $d \geq 3$. The design of our algorithm is based on the idea of path-decomposition Latin square and its relevant properties. In [13], Kim and Chung introduced a similar concept called Hamiltonian circuit Latin square for solving a parallel routing problem on RC-graphs. In that paper, a set of m internally disjoint paths on $G(2^m, 4)$ are constructed. A parallel routing problem on a network is a problem that looks for a set of routing paths for handling multiple data items simultaneously transmitted from a source node to a destination node. In order for all data items to arrive at the destination node quickly and securely, any two paths are demanded to be internally disjoint. It is undoubted from the definition that the ISTs problem is harder than the parallel routing problem. In fact, any algorithm that is developed for solving the ISTs problem on a network can be applied to solve the parallel routing problem, but the converse does not hold.

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